

LARGE-ANGLE STATISTICS OF THE CMB ANGULAR CORRELATION FUNCTION: TESTING Λ CDM AT LOW MULTIPOLES

ÁLVARO MÉNDEZ RODRÍGUEZ DE TEMBLEQUE

Università degli Studi di Padova, Department of Physics and Astronomy “Galileo Galilei”

YAGO ASCASIBAR SEQUEIROS

Universidad Autónoma de Madrid, Departamento de Física Teórica

Version July 4, 2026

ABSTRACT

We investigate the statistical properties of the cosmic microwave background (CMB) angular correlation function at large angular scales ($\ell \leq 30$) using Planck legacy data. Three complementary statistics are studied: the integrated squared correlation $S_{\theta_a}^{\theta_b}$, the interval-averaged correlation $\langle \xi \rangle_{\theta_a}^{\theta_b}$, and the antipodal correlation C_{180} . Later, the statistics are combined using the Aggregated Cauchy Association Test (ACAT) to quantify the overall significance of the statistics, mitigating the ‘look-elsewhere’ effect. Theoretical distributions are generated via two independent methods, sampling from the MCMC posterior of the standard flat Λ CDM model fitted with the CamSpec high-multipole TT likelihood, and Monte Carlo realizations around the best-fit spectrum, allowing a direct comparison of the two approaches. While a lack of power correlation is observed, we find no statistically significant evidence that the Λ CDM model fails to fit the correlation function when combining all the measured statistics.

Subject headings: cosmic microwave background, angular correlation function, large-angle anomalies, Planck, Λ CDM

1. INTRODUCTION

Since the first measurements of the CMB temperature anisotropies, the largest angular scales have attracted particular attention because they probe the longest wavelength perturbations accessible within the observable universe. Early analyses of the COBE-DMR data already suggested that the angular two-point correlation function was unexpectedly close to zero for separations larger than approximately 60° (Hinshaw et al. 1996). Although the statistical power of COBE was limited, this feature was subsequently confirmed with much higher precision by the first-year WMAP observations (Spergel et al. 2003).

Figure 1 illustrates this suppression directly for the primary CamSpec model: the observed correlation function falls below the theoretical $\pm 1\sigma$ band for separations $\theta \gtrsim 60^\circ$.

To quantify the apparent suppression of correlations on large angular scales, Spergel et al. introduced the statistic

$$S_{1/2} = \int_{-1}^{1/2} C(\theta)^2 d(\cos \theta),$$

which measures the integrated squared correlation function over angular separations larger than 60° (Spergel et al. 2003). The observed value of $S_{1/2}$ was found to lie in the extreme tail of the Λ CDM distribution gener-

ated from Monte Carlo realizations, establishing the lack of large-angle correlations as one of the most persistent large-scale anomalies in the CMB.

A series of studies by Copi, Huterer, Schwarz and Starkman strengthened this conclusion using successive WMAP data releases. In particular, analyses of cut-sky maps demonstrated that the observed correlation function is nearly zero for $\theta \gtrsim 60^\circ$, with probabilities typically below 10^{-3} under the standard cosmological model (Copi et al. 2006, 2009, 2013). These works emphasized that the anomaly cannot be explained simply by the low quadrupole amplitude alone, but instead reflects an unusual covariance structure among the lowest multipoles. They also noted connections with other large-scale anomalies, including the quadrupole-octopole alignment and hemispherical asymmetries.

The interpretation of the anomaly, however, remains controversial. Efstathiou, Ma and Hanson argued that estimators based directly on masked-sky correlation functions can exaggerate the discrepancy with Λ CDM, and showed that full-sky maximum-likelihood reconstructions yield results more consistent with theoretical expectations (Efstathiou et al. 2010). Extending this argument, Pontzen and Peiris considered a broader class of anisotropic Gaussian models and concluded that the observed correlation function does not constitute compelling evidence for new physics once the full space of possible covariance structures is taken into account (Pontzen & Peiris 2010). These studies highlighted the

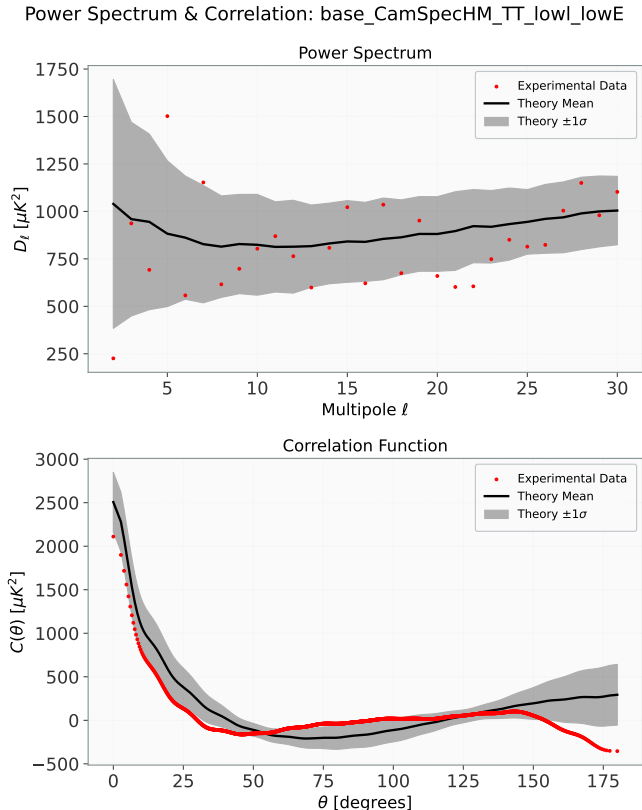


FIG. 1.— Power spectrum (top) and angular correlation function (bottom) for model `base_CamSpecHM_TT_lowl_lowE`. Experimental Planck data (red points), theoretical mean (black line), and $\pm 1\sigma$ band (grey) are shown. The suppression of $C(\theta)$ for $\theta \gtrsim 60^\circ$ relative to the theoretical expectation is visible in the lower panel.

importance of estimator choice, a posteriori statistic selection, and the definition of the theoretical ensemble used when assessing significance.

Another line of investigation has focused on the relationship between the lack of large-angle correlations and other low-multipole anomalies. Sarkar et al. tested whether the suppression of $C(\theta)$ is statistically connected to the well-known quadrupole-octopole alignments and found essentially no correlation between the two effects within Λ CDM realizations (Sarkar et al. 2010). The simultaneous occurrence of both anomalies is therefore substantially rarer than either anomaly considered individually. Similarly, Kim and Naselsky argued that the observed odd-parity preference at low multipoles provides a phenomenological explanation for the deficit of large-angle correlations, although no convincing physical origin was identified (Kim & Naselsky 2011).

The anomaly has also motivated the development of alternative statistical descriptions of large-scale correlations. Direction-dependent generalizations of the correlation function were introduced to investigate whether the missing power is localized on the sky rather than uniformly distributed. Zhang constructed directional versions of both $C(\theta)$ and $S_{1/2}$, showing that a large fraction of the full-sky signal originates from regions associated with the quadrupole and octopole extrema (Zhang 2012).

Other approaches include bipolar harmonic decompositions (Joshi et al. 2009), generalized isotropic angular correlation functions (Dipanshu et al. 2023), geometric representations of the two-point function (Froes et al. 2015), and all-angle statistics such as S_1 , proposed to reduce the dependence on the a posteriori choice of the 60° threshold (Gruppuso 2014).

The persistence of the anomaly in increasingly precise datasets has further motivated investigations beyond temperature anisotropies. Analyses of Planck polarization maps have revealed a similar suppression of large-angle correlations, with significances comparable to those found in temperature data (Chiocchetta et al. 2020). If confirmed, such results would suggest that the anomaly is not solely a consequence of temperature-specific systematics and may instead reflect a more fundamental property of the largest observable scales.

Despite nearly two decades of study, there is still no consensus regarding either the statistical significance or the physical interpretation of the lack of large-angle correlations. Part of this ambiguity arises from the existence of many partially correlated test statistics, each emphasizing different aspects of the low- ℓ temperature field. A rigorous assessment therefore requires both a careful construction of the theoretical sampling distribution and a framework capable of combining evidence from multiple correlated observables. To address this problem, in this study we combine the statistics using the ACAT test in order to measure the overall significance of the model: although a single statistic can be constructed to show greater significance in isolation, the overall significance is reduced once all statistics are combined, and therefore we obtain a more robust measure of the tension with Λ CDM.

Therefore, in this work we revisit the large-angle correlation function using Planck 2018 data, focusing on three complementary statistics: the integrated squared correlation $S_{\theta_a}^{\theta_b}$, the interval-averaged correlation $\langle \xi \rangle_{\theta_a}^{\theta_b}$, and the antipodal correlation C_{180} . We construct theoretical sampling distributions using both Monte Carlo realizations around the best-fit Λ CDM spectrum and samples drawn from the full Planck posterior, allowing us to disentangle cosmic variance from parameter uncertainty. Finally, we combine the individual p-values of all statistics using the Aggregated Cauchy Association Test (ACAT) to obtain a global measure of tension with Λ CDM.

2. STATISTICS

The statistics studied here are designed to probe complementary aspects of the large-angle correlation function: its integrated power, its mean level over specific angular ranges, and its value at antipodal separation. The choice is guided by the existing literature on low- ℓ anomalies, though as noted by Efstathiou et al. (2010), some statistics, including $S_{1/2}$, were identified a posteriori from the data, which inflates their apparent significance. Studying multiple statistics simultaneously, and combining their p-values through a global test, is therefore essential to obtain a reliable assessment of tension with Λ CDM.

2.1. Angular Power Spectrum and Correlation Function

The CMB temperature anisotropy $\Delta T(\hat{n})/T_0$ is expanded in spherical harmonics with coefficients $a_{\ell m}$. For a statistically isotropic Gaussian field the two-point statistics are fully encoded in the angular power spectrum $C_\ell = \langle |a_{\ell m}|^2 \rangle$. We work with the conventional scaled spectrum $D_\ell = \ell(\ell+1)C_\ell/(2\pi)$, which is approximately flat for a scale-invariant primordial spectrum.

The angular two-point correlation function is obtained by transforming back to real space via a Legendre expansion,

$$C(\theta) = \sum_{\ell} \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos \theta) = \sum_{\ell} \frac{2\ell+1}{2\ell(\ell+1)} D_\ell P_\ell(\cos \theta), \quad (1)$$

where P_ℓ are Legendre polynomials. The sum runs from $\ell = 2$ to $\ell_{\max} = 30$ throughout this work.

2.2. Interval Statistics

Three statistics are evaluated on each of the angular intervals listed in Table 1, chosen to partition the accessible range $[0^\circ, 180^\circ]$ into physically motivated regions while including the classic $S_{1/2}$ interval for comparison with the literature.

Antipodal correlation $C_{180} \equiv C(\theta = 180^\circ)$ measures the correlation between antipodal points on the sky. Because it is a pointwise evaluation of $C(\theta)$ it is relatively insensitive to masking effects at intermediate scales, and it provides a direct test of the near-vanishing of large-angle correlations at the largest possible separation.

Interval-averaged correlation $\langle \xi \rangle_{\theta_a}^{\theta_b}$ is the mean of $C(\theta)$ over the angular interval $[\theta_a, \theta_b]$, expressed in terms of $\mu = \cos \theta$ as

$$\langle \xi \rangle_{\theta_a}^{\theta_b} = \frac{1}{b-a} \int_a^b C(\theta) d\mu \quad (2)$$

$$= \sum_{\ell} \frac{D_\ell}{2\ell(\ell+1)} \frac{P_{\ell+1}(b) - P_{\ell-1}(b) - P_{\ell+1}(a) + P_{\ell-1}(a)}{b-a}, \quad (3)$$

where $a = \cos \theta_b$ and $b = \cos \theta_a$. Being linear in D_ℓ , this statistic is sensitive to shifts in the mean correlation level within an interval and changes sign when the correlation function crosses zero, making it complementary to $S_{\theta_a}^{\theta_b}$.

Integrated squared correlation $S_{\theta_a}^{\theta_b}$ is defined as

$$S_{\theta_a}^{\theta_b} = \int_a^b C^2(\theta) \sin \theta d\theta = \sum_{n,m} A_n A_m D_n D_m T_{nm}, \quad (4)$$

where $A_\ell = (2\ell+1)/[2\ell(\ell+1)]$ and

$$T_{nm} = \sum_{r=0}^{\min(m,n)} \frac{A_r A_{m-r} A_{n-r} [P_{m+n-2r+1} - P_{m+n-2r-1}]_a^b}{A_{m+n-r} (2m+2n-2r+1)}$$

are overlap integrals of Legendre polynomials over the interval (Carlitz 1961), a similar expression to the one obtained in (Copi et al. 2009), but extended to different angles. Being quadratic in D_ℓ , this statistic amplifies deviations that are coherent across multipoles and

TABLE 1: Angular intervals used for the statistics. The intervals were obtained by solving $C(\theta) = 0$ for the observed (Planck) correlation function.

Angular range	$[\cos \theta_b, \cos \theta_a]$
$[0^\circ, 30^\circ]$	$[0.865, 1.000]$
$[30^\circ, 92^\circ]$	$[-0.039, 0.865]$
$[92^\circ, 154^\circ]$	$[-0.901, -0.039]$
$[154^\circ, 180^\circ]$	$[-1.000, -0.901]$
$[60^\circ, 180^\circ]$	$[-1.000, 0.500]$

is always non-negative, so it cannot be cancelled by sign changes of $C(\theta)$ within the interval. The classic $S_{1/2}$ statistic of Spergel et al. (2003) corresponds to the choice $[\theta_a, \theta_b] = [60^\circ, 180^\circ]$.

The intervals considered are listed in Table 1, together with the classic $[60^\circ, 180^\circ]$ range used to compute $S_{1/2}$. Rather than partitioning $[0^\circ, 180^\circ]$ into fixed, evenly spaced angular bins, the boundaries were placed at the zeros of the correlation function measured from the observed Planck multipoles (Aghanim et al. 2020). A fixed, arbitrary binning would not be representative of the measured $C(\theta)$: it could mix angular ranges in which the data is predominantly positive with ranges in which it is predominantly negative, and the resulting cancellation within $\langle \xi \rangle_{\theta_a}^{\theta_b}$ would dilute any genuine mismatch between the observed and predicted correlation function. Partitioning at the observed zero-crossings instead guarantees that each interval corresponds to a single coherent sign region of the measured signal, so that $\langle \xi \rangle_{\theta_a}^{\theta_b}$ directly compares the predicted and observed mean level lobe by lobe, and $S_{\theta_a}^{\theta_b}$ probes the predicted and observed power within the same regions. Figure 2 shows the resulting partition together with both the observed and the model-predicted correlation function $\xi(\theta)$ (top panel) and its square $\xi^2(\theta)$ (bottom panel), illustrating that the fixed intervals track the sign changes of the data rather than those of the Λ CDM prediction. For each interval, the horizontal segments mark the Planck value (red) and the model median (black), and the vertical error bars give the model's 68% credible interval; the classic $[60^\circ, 180^\circ]$ interval used for $S_{1/2}$ in the literature is drawn dashed to distinguish it from the four data-driven intervals, which are drawn solid.

We note that defining the intervals from a feature of the data under test is itself a form of a posteriori choice, of the kind discussed by Efstathiou et al. (2010) (Section 1). This affects only the two interval-based statistics, $S_{\theta_a}^{\theta_b}$ and $\langle \xi \rangle_{\theta_a}^{\theta_b}$; the antipodal statistic C_{180} , which yields the strongest individual tension found in this work (Tables 2 and 3), does not depend on this choice at all. The theoretical sampling distributions of Section 3.3 are evaluated on exactly the same intervals, so the reported p-values remain valid tests of whether the observed value is unusual within each interval; what they do not capture is the additional, harder-to-calibrate look-elsewhere effect of having selected the interval boundaries from the same dataset being tested. We mitigate this only partly, by retaining the classic $[60^\circ, 180^\circ]$ interval for direct comparison with the literature and by combining all eleven statistics through ACAT rather than reporting any single interval-based result in isolation.

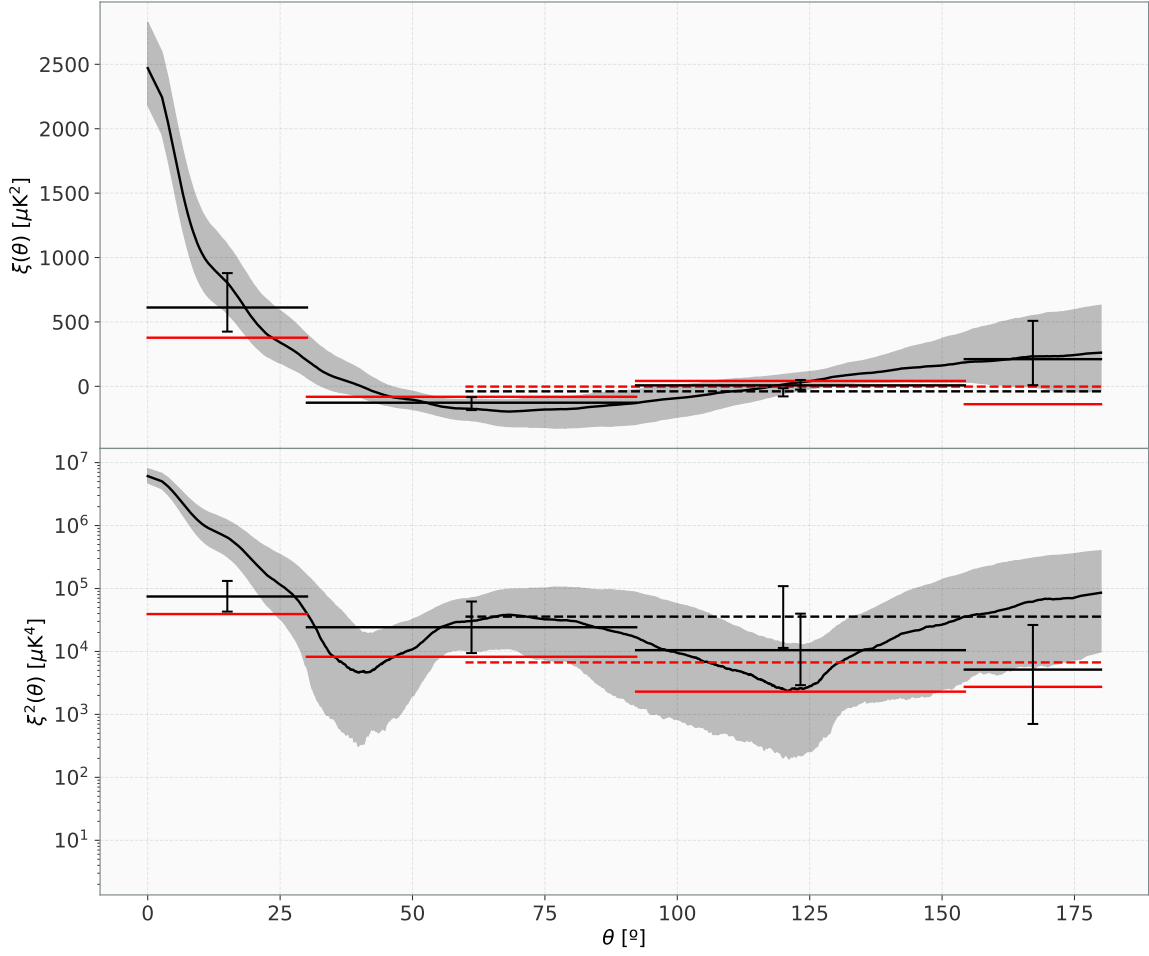


FIG. 2.— Correlation function $\xi(\theta)$ (top) and its square $\xi^2(\theta)$ (bottom, log scale) for model `base_CamSpecHM_TT_lowl_lowE`, showing the median and 68% band of the theoretical ensemble (black line and grey band) together with the per-interval $\langle \xi \rangle_{\theta_a}^{\theta_b}$ (top) and $S_{\theta_a}^{\theta_b}$ (bottom) diagnostics across the partitioned angular intervals of Table 1. Red segments mark the observed Planck value and black segments the model median in each interval, with vertical error bars showing the model's 68% credible interval; the classic $[60^\circ, 180^\circ]$ interval is dashed, the four data-driven intervals are solid.

3. DATA

3.1. Planck Posterior Chains

We use the publicly available Planck 2018 MCMC chains from the Planck Legacy Archive¹ (Planck Collaboration 2015). The primary dataset is the CamSpec high-multipole TT likelihood combined with the low- ℓ temperature and polarization likelihoods (base_CamSpecHM_TT_lowl_lowE). This provides posterior samples of the six standard Λ CDM parameters $\{\Omega_b h^2, \Omega_c h^2, H_0, \tau, n_s, A_s\}$ along with a suite of derived quantities. The Plik-based TT likelihood and two curvature-extended runs are considered in Appendix B.

For each posterior sample the theoretical power spectrum D_ℓ is computed with CAMB (Lewis ???), and the corresponding $C(\theta)$ and derived statistics are evaluated analytically using Equations 1–4.

3.2. Observed Planck Multipoles

The experimental values of each statistic are computed from the Planck 2018 Commander temperature map and associated beam and pixel window functions, using the same multipole range $\ell \in [2, 30]$. These observed values serve as the reference against which the theoretical distributions are compared.

3.3. Ensemble Construction

Two independent strategies are used to construct the theoretical sampling distribution. The full analysis pipeline is publicly available [here](#). The exact numerical results reported in this paper can be reproduced from the uploaded output files; because the pipeline uses a random seed for each realization, re-running it from scratch will reproduce the same statistical conclusions but not bit-identical histograms.

1. **MCMC ensemble.** For each of the 1000 samples drawn from the tail of the Planck posterior chain, specifically the last 1000 rows after GetDist’s default burn-in removal, the cosmological parameters are passed to CAMB to compute a theoretical spectrum D_ℓ^{th} . Each spectrum is then passed through the simulation pipeline described below, producing one realization of cosmic variance on top of the parameter-determined D_ℓ^{th} . The resulting ensemble therefore captures both parameter uncertainty and cosmic variance simultaneously, constituting a Bayesian predictive distribution for each statistic.
2. **Best-fit ensemble.** The best-fit D_ℓ^{bf} is read from the `.minimum.theory.c1` file of the Planck chain, which contains the spectrum evaluated at the maximum-posterior parameter point. This single spectrum is used as input to 1000 independent pipeline realizations, each with a different random seed. The spread of the resulting ensemble is driven entirely by cosmic variance at fixed cosmological parameters, with no contribution from parameter uncertainty. The width of its histograms is therefore a lower bound on the total variance: the additional width seen in the MCMC histograms is attributable purely to parameter uncertainty, and

comparing the two ensembles provides a clean decomposition of the two contributions.

In both cases, the pipeline follows the methodology of Sullivan et al. (2024). For each theoretical C_ℓ^{th} spectrum, the beam transfer function b_ℓ and the HEALPix pixel window function p_ℓ are applied, a full-sky temperature map is synthesized at $N_{\text{side}} = 2048$ using `hp.synfast`, and the pseudo- C_ℓ are recovered with `hp.anafast`. The recovered spectrum is then deconvolved by $b_\ell^2 p_\ell^2$ to obtain an estimate $\hat{D}_\ell$ in the same space as the published Planck values. When a sky mask is applied, the recovered C_ℓ are corrected by the sky fraction f_{sky} before deconvolution. At $\ell_{\text{max}} = 30$ the beam and pixel window corrections are numerically negligible, but they are included for consistency with the Planck data reduction.

In order to reduce the look-elsewhere effect and obtain a calibrated global significance that accounts for the correlation structure of the statistics without requiring an explicit model of it, we use the Aggregated Cauchy Association Test (ACAT; Liu & Xie 2020). The eleven statistics examined, five intervals each for $S_{\theta_a}^{\theta_b}$ and $\langle \xi \rangle_{\theta_a}^{\theta_b}$, plus C_{180} , are not independent: all derive from the same realization of D_ℓ , so their p-values are correlated by construction. Treating them as independent and applying a simple Bonferroni correction would be overly conservative, while reporting only the most extreme individual p-value would be anti-conservative.

For each statistic $k \in \{1, \dots, 11\}$ the individual p-value p_k is computed as the one-sided empirical quantile of the observed value within the simulated ensemble: $p_k = F_k$, where F_k is the fraction of realizations with a statistic value less than or equal to the observed one. The ACAT test statistic is then

$$T = \sum_{k=1}^{11} w_k \tan\left[\left(\frac{1}{2} - p_k\right) \pi\right],$$

where equal weights $w_k = 1/11$ are used, reflecting no a priori preference for any particular angular interval or statistic. Under the null hypothesis, T follows a standard Cauchy distribution regardless of the dependence structure among the p_k , so the combined p-value is obtained analytically as

$$p_{\text{global}} = \frac{1}{2} - \frac{\arctan T}{\pi}.$$

The Cauchy combination is exact under arbitrary dependence, including the strong correlations expected between statistics computed over overlapping angular ranges. The global tension in units of Gaussian standard deviations is reported as $n_\sigma = \Phi^{-1}(1 - p_{\text{global}}/2)$, where Φ^{-1} is the probit function.

4. RESULTS

4.1. Best-Fit Analysis

In the best-fit analysis the theoretical sampling distribution is generated by Monte Carlo realizations of the CMB sky drawn from the best-fit CamSpec Λ CDM spectrum. This approach isolates cosmic variance from parameter uncertainty.

Figure 3 displays the distributions of the $\langle \xi \rangle_{\theta_a}^{\theta_b}$ (left column) and $S_{\theta_a}^{\theta_b}$ (right column) statistics, for all five

¹ <https://pla.esac.esa.int/>

angular intervals. Each panel shows the distribution over the 1000 Monte Carlo realizations (blue histogram), and the observed Planck value (red vertical line). Figure 4 shows the analogous distribution for C_{180} .

In that set of histograms we can see that the observed values of $\langle \xi \rangle_{\theta_a}^{\theta_b}$ sometimes are lower and sometimes higher than the median of the distribution, which is consistent with the fact that $\langle \xi \rangle_{\theta_a}^{\theta_b}$ is a linear statistic, and therefore it is more sensitive to shifts in the mean correlation level, and the change of sign in the correlation function in certain intervals.

As we can see in Table 2, for the different intervals all values of $S_{\theta_a}^{\theta_b}$ are below the median of the distribution, which is not only consistent with the lack of large-angle correlations but also suggests that the lack is not only present at large angles but also at intermediate and small angles.

The statistical summary, including the median, 68% credible interval, observed value, and empirical p-value for each statistic, is reported in Table 2.

4.2. MCMC Analysis

In the MCMC analysis the theoretical sampling distribution is constructed directly from the Planck posterior, propagating full parameter uncertainty through to the derived statistics. The 1000 posterior samples drawn from the CamSpec Λ CDM chain each yield a distinct D_ℓ spectrum, whose distribution reflects both cosmic variance and parameter degeneracies.

Figure 5 shows the resulting distributions of $\langle \xi \rangle_{\theta_a}^{\theta_b}$ (left column) and $S_{\theta_a}^{\theta_b}$ (right column), and Figure 6 shows the analogous distribution for C_{180} . The broader width of the MCMC-derived histograms compared to the best-fit case reflects the additional spread due to cosmological parameter uncertainty.

We see the same trend as in the best-fit analysis, with the observed values of $S_{\theta_a}^{\theta_b}$ consistently below the median of the distribution. The overall tension with Λ CDM is slightly increased in the MCMC analysis, mainly due to the additional contribution of parameter uncertainty, which broadens the theoretical distribution and shifts the median slightly.

The statistical summary, including the median, 68% credible interval, observed value, and empirical p-value

for each statistic, is reported in Table 3.

5. DISCUSSION

The results for alternative likelihoods and models with free curvature Ω_k are reported in Appendix B. The dependence of the results on the likelihood choice (CamSpec vs. Plik) provides a useful cross-check, while the curvature-extended models test whether a departure from flatness could alleviate the observed tensions.

As noted by (Kim & Naselsky 2011), a lack of correlation is observed across all angular scales—including large, intermediate, and small angles. This is reflected in the fact that the observed values of $S_{\theta_a}^{\theta_b}$ fall below the distribution’s median for all intervals. This behavior is observed across all models and likelihoods examined, as detailed in Appendix B. While the significance of the individual statistics is not particularly high, the consistent pattern of low $S_{\theta_a}^{\theta_b}$ values across all intervals suggests a coherent suppression of correlations that is not confined to a flat model, but rather extends to models with free curvature.

6. CONCLUSIONS

We have studied the large-angle CMB angular correlation function and its derived statistics using Planck 2018 data and two complementary ensembles for the theoretical sampling distributions. The statistics examined, the quadratic $S_{\theta_a}^{\theta_b}$, the linear $\langle \xi \rangle_{\theta_a}^{\theta_b}$, and the antipodal C_{180} , probe distinct aspects of the large-angle temperature field.

Both ensembles provide complementary perspectives on the significance of the large-angle anomalies. The best-fit ensemble isolates the contribution of cosmic variance, while the MCMC ensemble captures the full parameter uncertainty and the cosmic variance.

The large-angle suppression of the correlation function, most evident in C_{180} , is reproducible across both ensembles and across all likelihood and curvature variants tested (Appendix B), but it is not statistically significant once the eleven correlated statistics are combined through the ACAT global test: the resulting tension with Λ CDM is $\sim 1.1\sigma$ for the primary `base_CamSpecHM.TT_lowl_lowE` model (Tables 2 and 3).

We conclude that current Planck data do not provide significant evidence against the standard cosmological model from the large-angle correlation function.

REFERENCES

- Aghanim, N., Akrami, Y., Ashdown, M., et al. 2020, *Astronomy and Astrophysics*, 641, A5, doi: [10.1051/0004-6361/201936386](https://doi.org/10.1051/0004-6361/201936386)
- Carlitz, L. 1961, *Archiv der Mathematik*, 12, 334, doi: [10.1007/BF01650571](https://doi.org/10.1007/BF01650571)
- Chiocchetta, C., Gruppuso, A., Lattanzi, M., Natoli, P., & Pagano, L. 2020, *Journal of Cosmology and Astroparticle Physics*, 2021, doi: [10.1088/1475-7516/2021/08/015](https://doi.org/10.1088/1475-7516/2021/08/015)
- Copi, C., Huterer, D., Schwarz, D., & Starkman, G. 2006, *Physical Review D*, 75, 023507, doi: [10.1103/physrevd.75.023507](https://doi.org/10.1103/physrevd.75.023507)
- . 2013, *Monthly Notices of the Royal Astronomical Society*, 451, 2978, doi: [10.1093/mnras/stv1143](https://doi.org/10.1093/mnras/stv1143)
- Copi, C. J., Huterer, D., Schwarz, D. J., & Starkman, G. D. 2009, *Monthly Notices of the Royal Astronomical Society*, 399, 295–303, doi: [10.1111/j.1365-2966.2009.15270.x](https://doi.org/10.1111/j.1365-2966.2009.15270.x)
- Dipanshu, Souradeep, T., & Hirve, S. 2023, *The Astrophysical Journal*, 954, doi: [10.3847/1538-4357/ace895](https://doi.org/10.3847/1538-4357/ace895)
- Efstathiou, G., Ma, Y.-Z., & Hanson, D. 2010, *Monthly Notices of the Royal Astronomical Society*, 407, 2530, doi: [10.1111/j.1365-2966.2010.17081.x](https://doi.org/10.1111/j.1365-2966.2010.17081.x)
- Froes, A. L. D., Pereira, T. S., Bernui, A., & Starkman, G. 2015, *Physical Review D*, 92, doi: [10.1103/physrevd.92.043508](https://doi.org/10.1103/physrevd.92.043508)
- Gruppuso, A. 2014, *Monthly Notices of the Royal Astronomical Society*, 437, 2076, doi: [10.1093/mnras/stt1937](https://doi.org/10.1093/mnras/stt1937)
- Hinshaw, G., Banday, A. J., Bennett, C. L., et al. 1996, *The Astrophysical Journal*, 464, L25–L28, doi: [10.1086/310076](https://doi.org/10.1086/310076)
- Joshi, N., Jhingan, S., Souradeep, T., & Hajian, A. 2009, *Physical Review D*, 81, 083012, doi: [10.1103/physrevd.81.083012](https://doi.org/10.1103/physrevd.81.083012)
- Kim, J., & Naselsky, P. 2011, *The Astrophysical Journal*, 739, 79, doi: [10.1088/0004-637X/739/2/79](https://doi.org/10.1088/0004-637X/739/2/79)
- Lewis, A. ????, *CAMB: Code for Anisotropies in the Microwave Background*, <https://github.com/cmbant/CAMB>
- Liu, Y., & Xie, J. 2020, *Journal of the American Statistical Association*, 115, 393, doi: [10.1080/01621459.2018.1554485](https://doi.org/10.1080/01621459.2018.1554485)
- Planck Collaboration. 2015, *Planck Legacy Archive*, <https://pla.esac.esa.int/>

TABLE 2: Statistical summary for the best-fit analysis of model `base.CamSpecHM-TT_lowl_lowE`. Columns give the experimental value, best-fit median, 68% interval, and empirical one sided p-value.

Global tension (ACAT): $T = 2.091$, $p = 0.142$ (1.07σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	260.7976	[-38.2572, 629.7045]	0.022 (2.01σ)
$S_{180,60}$	6694.0882	35725.0265	[11308.2691, 108785.4640]	0.073 (1.45σ)
$S_{180,154}$	2739.9435	5144.2773	[704.4897, 26362.7403]	0.358 (0.36σ)
$S_{154,92}$	2298.0174	10463.2839	[2893.9033, 39807.8958]	0.133 (1.11σ)
$S_{92,30}$	8229.1759	24227.7116	[9406.5535, 62090.4792]	0.127 (1.14σ)
$S_{30,0}$	39257.1300	74487.4419	[42884.9866, 131879.2429]	0.114 (1.21σ)
$\xi_{180,60}$	-2.1052	-38.8758	[-78.4081, -14.6241]	0.966 (-1.83σ)
$\xi_{180,154}$	-138.6536	210.8664	[7.7966, 508.4919]	0.051 (1.64σ)
$\xi_{154,92}$	42.0072	7.6378	[-25.2907, 49.3801]	0.797 (-0.83σ)
$\xi_{92,30}$	-81.0404	-127.2322	[-184.4908, -84.2408]	0.862 (-1.09σ)
$\xi_{30,0}$	377.5806	611.8525	[424.7966, 878.9550]	0.098 (1.29σ)

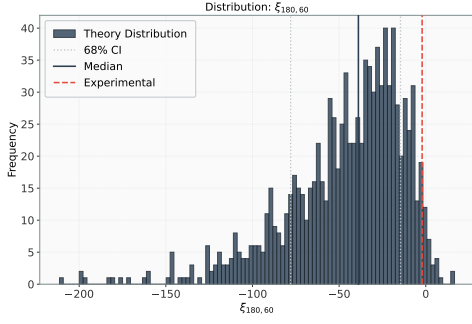
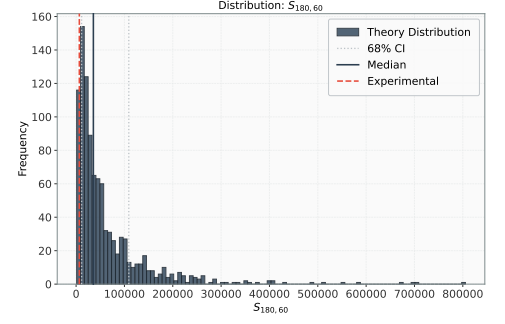
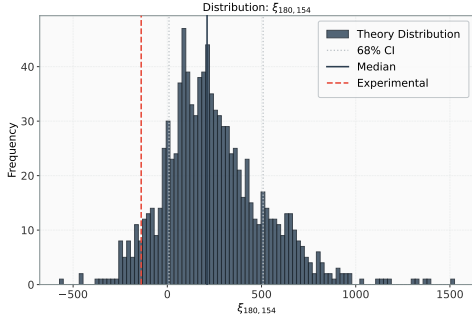
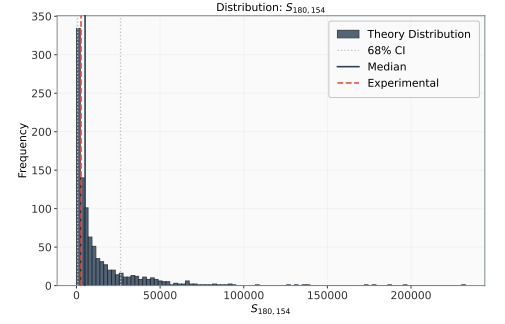
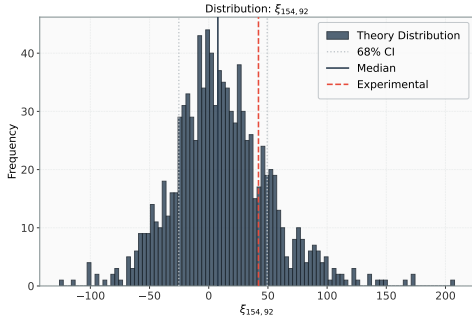
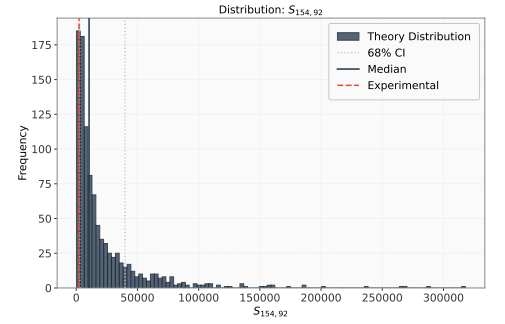
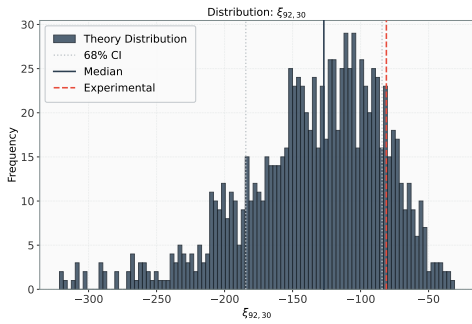
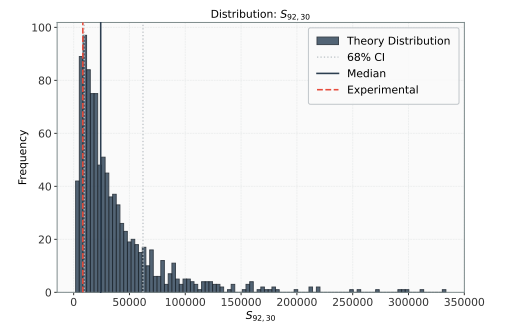
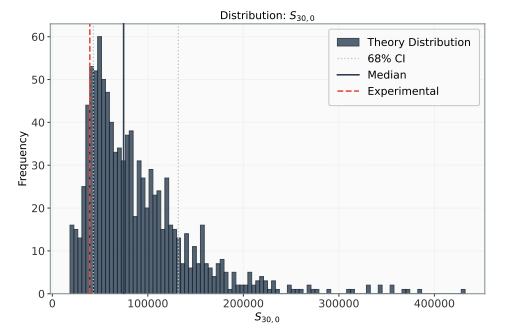
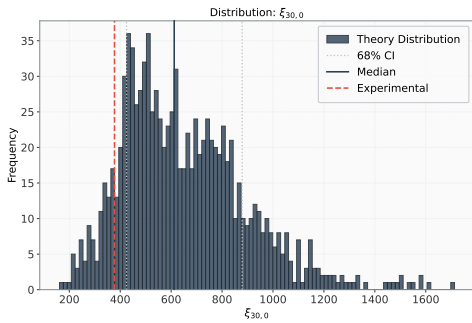
TABLE 3: Statistical summary for the MCMC analysis of model `base.CamSpecHM-TT_lowl_lowE`. Columns give the experimental value, MCMC median, 68% credible interval, and empirical one sided p-value.

Global tension (ACAT): $T = 2.348$, $p = 0.128$ (1.14σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	261.6952	[-39.0156, 589.9108]	0.018 (2.10σ)
$S_{180,60}$	6694.0882	33937.6161	[10094.0365, 102450.9702]	0.076 (1.43σ)
$S_{180,154}$	2739.9435	5200.7872	[664.4140, 23292.6452]	0.368 (0.34σ)
$S_{154,92}$	2298.0174	10088.0224	[2572.3400, 34735.6347]	0.138 (1.09σ)
$S_{92,30}$	8229.1759	23088.7680	[9244.2759, 60281.6008]	0.137 (1.09σ)
$S_{30,0}$	39257.1300	72409.8155	[41887.8294, 128755.4170]	0.133 (1.11σ)
$\xi_{180,60}$	-2.1052	-37.1448	[-77.0816, -13.6870]	0.962 (-1.77σ)
$\xi_{180,154}$	-138.6536	210.0608	[-7.9407, 473.6089]	0.047 (1.67σ)
$\xi_{154,92}$	42.0072	8.7805	[-22.1704, 48.5205]	0.802 (-0.85σ)
$\xi_{92,30}$	-81.0404	-126.3688	[-182.5317, -81.9472]	0.844 (-1.01σ)
$\xi_{30,0}$	377.5806	603.5009	[414.7549, 861.6303]	0.116 (1.20σ)

- Pontzen, A., & Peiris, H. V. 2010, Physical Review D, 81,
doi: [10.1103/physrevd.81.103008](https://doi.org/10.1103/physrevd.81.103008)
- Sarkar, D., Huterer, D., Copi, C., Starkman, G., & Schwarz, D.
2010, Astroparticle Physics, 34, 591,
doi: [10.1016/j.astropartphys.2010.12.009](https://doi.org/10.1016/j.astropartphys.2010.12.009)
- Spergel, D. N., Verde, L., Peiris, H. V., et al. 2003, The
Astrophysical Journal Supplement Series, 148, 175,
doi: [10.1086/377226](https://doi.org/10.1086/377226)
- Sullivan, R. M., Hergt, L. T., & Scott, D. 2024, Methods for
CMB map analysis. <https://arxiv.org/abs/2410.12951>
- Zhang, S. 2012, The Astrophysical Journal, 748, L20,
doi: [10.1088/2041-8205/748/2/L20](https://doi.org/10.1088/2041-8205/748/2/L20)

APPENDIX

A. MONTE CARLO HISTOGRAMS

(a) $\langle \xi \rangle_{60}^{180}$ (f) S_{60}^{180} (classic $S_{1/2}$)(b) $\langle \xi \rangle_{154}^{180}$ (g) S_{154}^{180} (c) $\langle \xi \rangle_{92}^{154}$ (h) S_{92}^{154} (d) $\langle \xi \rangle_{30}^{92}$ (i) S_{30}^{92} 

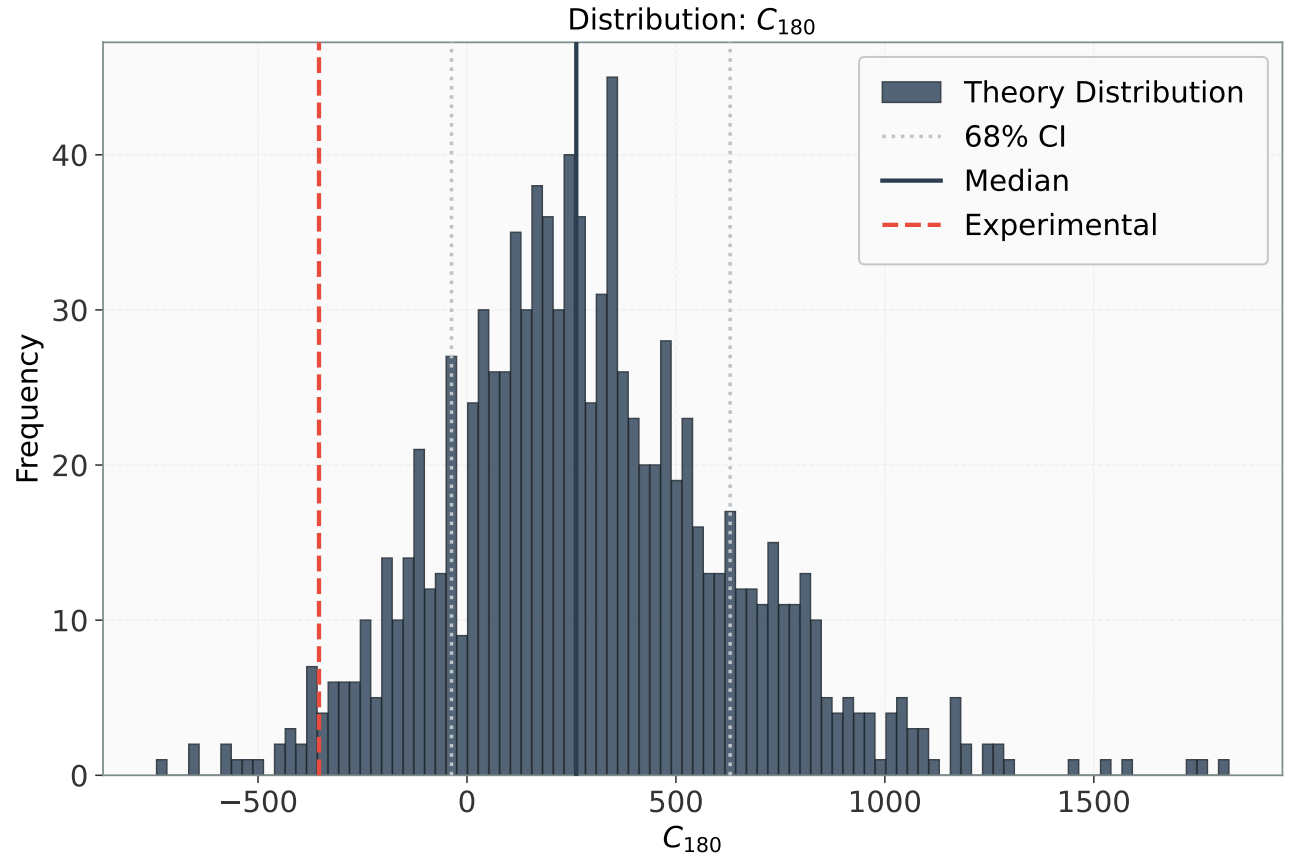
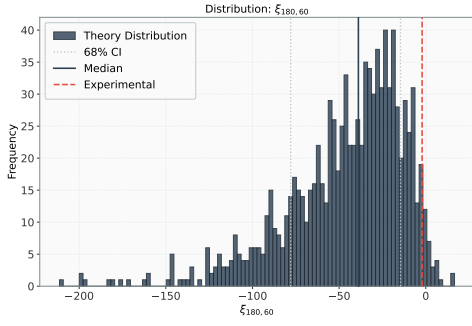
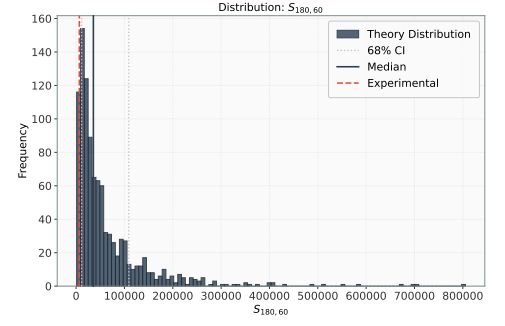
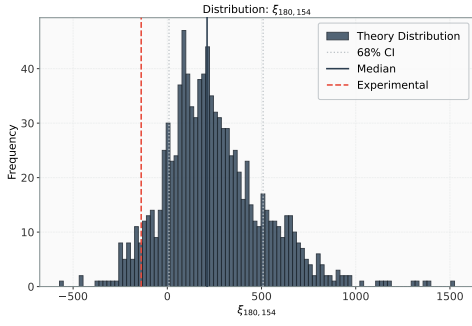
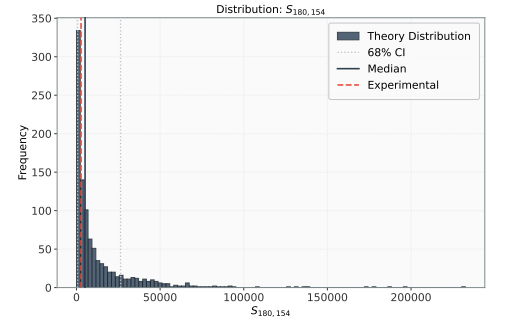
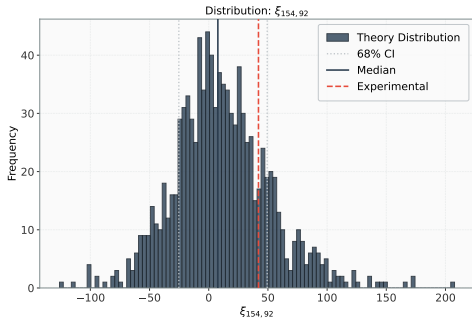
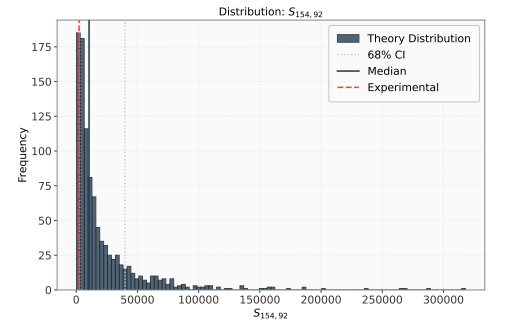
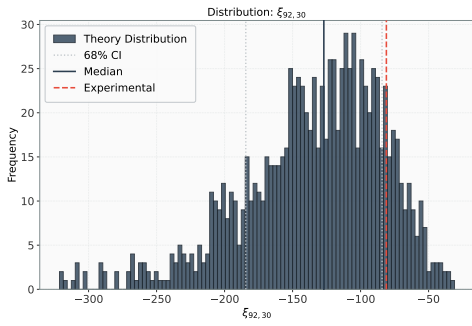
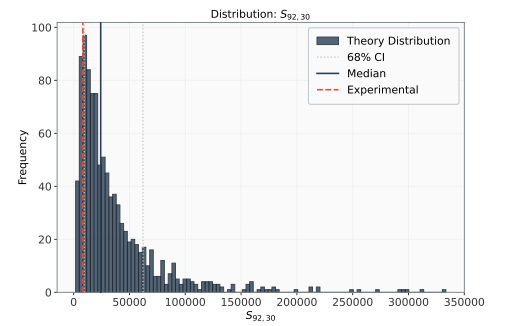
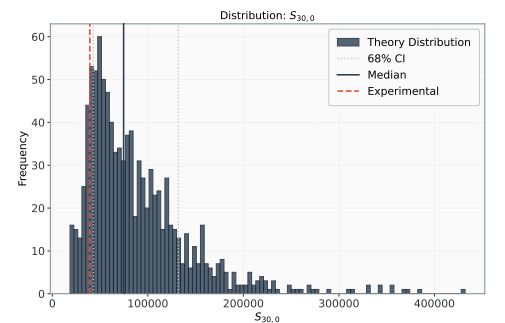
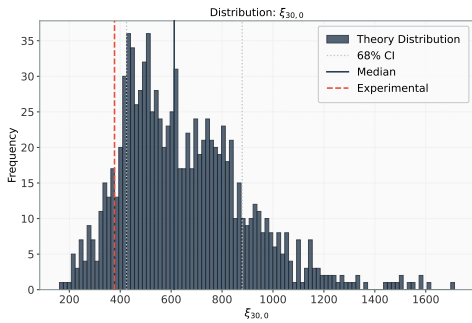


FIG. 4.—: Best-fit Monte Carlo distribution of the antipodal correlation C_{180} for model `base_CamSpecHM_TT_lowl_lowE`.

(a) $\langle \xi \rangle_{60}^{180}$ (f) S_{60}^{180} (classic $S_{1/2}$)(b) $\langle \xi \rangle_{154}^{180}$ (g) S_{154}^{180} (c) $\langle \xi \rangle_{92}^{154}$ (h) S_{92}^{154} (d) $\langle \xi \rangle_{30}^{92}$ (i) S_{30}^{92} 

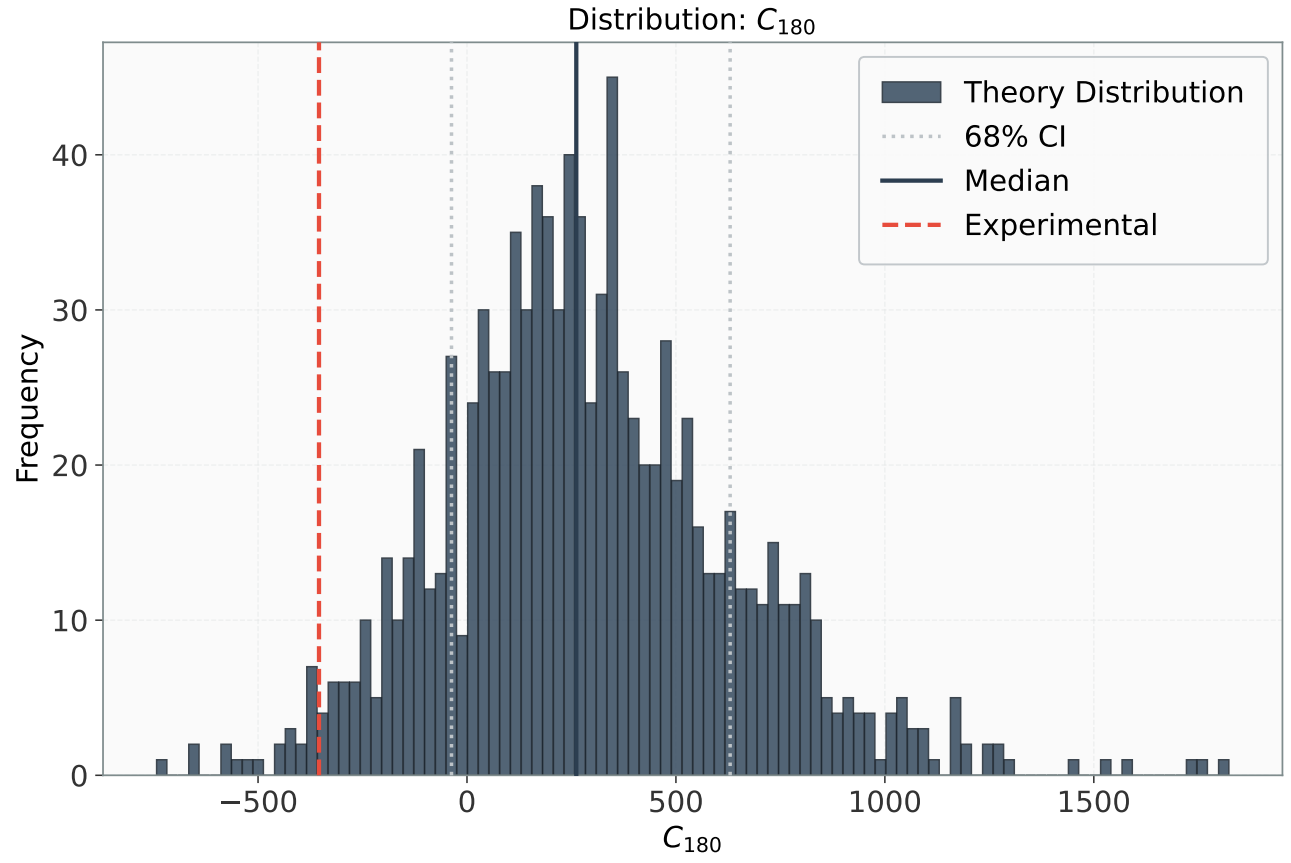


FIG. 6.—: MCMC posterior distribution of C_{180} for model `base_CamSpecHM_TT_lowl_lowE`.

B. ADDITIONAL MODELS: STATISTICAL TABLES

The tables in this appendix report the full statistical summary for three additional Planck models. All quantities are defined as in the main text. Model names follow the Planck convention: **omegak** denotes a run with free spatial curvature; **plikHM** denotes the Plik high-multipole likelihood; **TTTEEE** denotes a joint temperature and polarization fit.

B.1. MCMC Results

TABLE 4: MCMC results for **base_plikHM_TT_lowl_lowE**.

Global tension (ACAT): $T = 1.884$, $p = 0.155$ (1.01σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	241.2170	[-55.1699, 594.4383]	0.029 (1.90σ)
$S_{180,60}$	6694.0882	33801.8449	[11110.7265, 102182.3499]	0.065 (1.51σ)
$S_{180,154}$	2739.9435	4766.0801	[672.8630, 22097.9814]	0.376 (0.32σ)
$S_{154,92}$	2298.0174	10527.3578	[2848.7506, 34955.3646]	0.125 (1.15σ)
$S_{92,30}$	8229.1759	23310.8042	[9398.8164, 59071.8598]	0.128 (1.14σ)
$S_{30,0}$	39257.1300	73229.9679	[42537.9015, 125384.8849]	0.126 (1.15σ)
$\xi_{180,60}$	-2.1052	-38.5341	[-76.0073, -14.4676]	0.962 (-1.77σ)
$\xi_{180,154}$	-138.6536	202.3885	[-6.9888, 468.8826]	0.052 (1.63σ)
$\xi_{154,92}$	42.0072	9.1633	[-25.4628, 51.1022]	0.781 (-0.78σ)
$\xi_{92,30}$	-81.0404	-126.3941	[-178.6658, -84.8224]	0.860 (-1.08σ)
$\xi_{30,0}$	377.5806	607.6992	[416.0714, 851.9444]	0.092 (1.33σ)

TABLE 5: MCMC results for **base_omegak_CamSpecHM_TT_lowl_lowE**.

Global tension (ACAT): $T = 3.250$, $p = 0.095$ (1.31σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	208.9942	[-34.4807, 486.2964]	0.011 (2.29σ)
$S_{180,60}$	6694.0882	21765.8299	[8109.5135, 74119.3517]	0.119 (1.18σ)
$S_{180,154}$	2739.9435	3267.8897	[391.6000, 15964.8461]	0.466 (0.09σ)
$S_{154,92}$	2298.0174	7021.0369	[2079.4052, 26823.1748]	0.186 (0.89σ)
$S_{92,30}$	8229.1759	15837.8991	[6161.9693, 41989.8915]	0.246 (0.69σ)
$S_{30,0}$	39257.1300	51438.9870	[29716.1148, 94216.3869]	0.310 (0.50σ)
$\xi_{180,60}$	-2.1052	-30.7119	[-64.0166, -12.3069]	0.957 (-1.72σ)
$\xi_{180,154}$	-138.6536	164.2943	[-0.5697, 396.1609]	0.030 (1.88σ)
$\xi_{154,92}$	42.0072	6.7193	[-20.7478, 41.0697]	0.847 (-1.02σ)
$\xi_{92,30}$	-81.0404	-105.5266	[-152.8439, -67.5677]	0.734 (-0.62σ)
$\xi_{30,0}$	377.5806	502.4040	[340.1959, 725.3792]	0.232 (0.73σ)

TABLE 6: MCMC results for **base_omegak_plikHM_TTTEEE_lowl_lowE**.

Global tension (ACAT): $T = 2.615$, $p = 0.116$ (1.19σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	208.6258	[-45.5817, 506.9834]	0.013 (2.23σ)
$S_{180,60}$	6694.0882	25836.5779	[8014.3153, 74173.0515]	0.111 (1.22σ)
$S_{180,154}$	2739.9435	3617.6754	[464.5482, 16839.1825]	0.455 (0.11σ)
$S_{154,92}$	2298.0174	7418.2542	[2223.0737, 24954.1641]	0.172 (0.95σ)
$S_{92,30}$	8229.1759	18231.4590	[6845.5271, 42367.6853]	0.205 (0.82σ)
$S_{30,0}$	39257.1300	55436.6264	[32260.9180, 96229.5173]	0.264 (0.63σ)
$\xi_{180,60}$	-2.1052	-33.4438	[-63.2413, -11.6939]	0.961 (-1.76σ)
$\xi_{180,154}$	-138.6536	168.6679	[-15.5933, 403.9776]	0.040 (1.75σ)
$\xi_{154,92}$	42.0072	9.8408	[-20.0036, 45.5296]	0.816 (-0.90σ)
$\xi_{92,30}$	-81.0404	-110.8947	[-155.6151, -70.9814]	0.766 (-0.73σ)
$\xi_{30,0}$	377.5806	522.1450	[353.3445, 736.3491]	0.196 (0.86σ)

B.2. Best-Fit Results

This paper was built using the Open Journal of Astrophysics L^AT_EX template. The OJA is a journal which provides fast and easy peer review for new papers in the **astro-ph** section of the arXiv, making the reviewing process simpler for authors and referees alike. Learn more at <http://astro.theoj.org>.

TABLE 7: Best-fit results for `base_plikHM.TT_lowl_lowE`.

Global tension (ACAT): $T = 1.753$, $p = 0.165$ (0.97σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	235.3880	[-62.2566, 586.2191]	0.028 (1.91σ)
$S_{180,60}$	6694.0882	33261.4972	[10608.9403, 100001.9703]	0.082 (1.39σ)
$S_{180,154}$	2739.9435	4638.8774	[579.8045, 21964.4118]	0.379 (0.31σ)
$S_{154,92}$	2298.0174	9746.7383	[2727.4077, 35795.7064]	0.130 (1.13σ)
$S_{92,30}$	8229.1759	23100.4791	[9000.3718, 60298.5831]	0.145 (1.06σ)
$S_{30,0}$	39257.1300	70442.7090	[40918.0369, 130962.8091]	0.137 (1.09σ)
$\xi_{180,60}$	-2.1052	-37.0548	[-75.1144, -13.0243]	0.959 (-1.74σ)
$\xi_{180,154}$	-138.6536	195.4772	[-25.0376, 460.2276]	0.061 (1.55σ)
$\xi_{154,92}$	42.0072	10.7736	[-22.3436, 56.5905]	0.774 (-0.75σ)
$\xi_{92,30}$	-81.0404	-124.4639	[-185.5235, -80.1492]	0.835 (-0.97σ)
$\xi_{30,0}$	377.5806	595.2820	[409.3589, 866.2165]	0.113 (1.21σ)

TABLE 8: Best-fit results for `base_omegak_CamSpecHM.TT_lowl_lowE`.

Global tension (ACAT): $T = 2.686$, $p = 0.113$ (1.21σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	211.9556	[-29.8411, 502.2334]	0.014 (2.20σ)
$S_{180,60}$	6694.0882	23719.2671	[7675.5717, 73031.1322]	0.131 (1.12σ)
$S_{180,154}$	2739.9435	3625.1233	[406.7749, 15623.4577]	0.453 (0.12σ)
$S_{154,92}$	2298.0174	7128.0973	[1943.9653, 24772.1131]	0.209 (0.81σ)
$S_{92,30}$	8229.1759	16340.3031	[6726.9125, 41856.6421]	0.223 (0.76σ)
$S_{30,0}$	39257.1300	51705.4424	[31338.3396, 91825.7121]	0.295 (0.54σ)
$\xi_{180,60}$	-2.1052	-32.3840	[-63.4182, -11.6872]	0.953 (-1.67σ)
$\xi_{180,154}$	-138.6536	172.3367	[-0.9287, 394.2349]	0.034 (1.83σ)
$\xi_{154,92}$	42.0072	8.3009	[-20.7864, 42.4205]	0.835 (-0.97σ)
$\xi_{92,30}$	-81.0404	-105.6599	[-151.8632, -71.0518]	0.747 (-0.67σ)
$\xi_{30,0}$	377.5806	504.0592	[354.5585, 720.7156]	0.211 (0.80σ)

TABLE 9: Best-fit results for `base_omegak_plikHM.TTTEEE_lowl_lowE`.

Global tension (ACAT): $T = 5.786$, $p = 0.054$ (1.60σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	218.3127	[-17.0392, 508.2853]	0.006 (2.51σ)
$S_{180,60}$	6694.0882	24349.6721	[7615.2117, 72133.8665]	0.136 (1.10σ)
$S_{180,154}$	2739.9435	3289.8516	[501.4208, 16347.5457]	0.459 (0.10σ)
$S_{154,92}$	2298.0174	7095.9810	[2120.8351, 26428.7002]	0.171 (0.95σ)
$S_{92,30}$	8229.1759	17049.6488	[6261.6264, 42388.9107]	0.224 (0.76σ)
$S_{30,0}$	39257.1300	54447.2715	[31086.6010, 96125.6258]	0.280 (0.58σ)
$\xi_{180,60}$	-2.1052	-33.5933	[-63.5443, -11.2915]	0.967 (-1.84σ)
$\xi_{180,154}$	-138.6536	172.7077	[9.0572, 400.9675]	0.020 (2.05σ)
$\xi_{154,92}$	42.0072	7.2581	[-21.0722, 40.1496]	0.853 (-1.05σ)
$\xi_{92,30}$	-81.0404	-106.5318	[-154.6407, -69.6381]	0.744 (-0.66σ)
$\xi_{30,0}$	377.5806	518.2218	[349.5821, 732.1247]	0.211 (0.80σ)